

AVIV Data Platform & Analytics Engineering Case

October 2025

1. Context

At AVIV Group (parent company of [SeLogger.com](https://www.seloger.com)), our mission is to simplify how people find, sell, and rent homes.

Behind the scenes, our data platform (built on AWS, Snowflake, and dbt) integrates property listings, leads, and marketing data to power analytics and decision-making.

You're joining the Data Platform team as a Data / Analytics Engineer.

Your challenge is to show how you would turn raw SeLogger-like data into a trusted, business-ready dataset that delivers value.

2. The Challenge

You'll work with small datasets that represent SeLogger's core business:

- Property Listings
 - listing_id
 - property_type (e.g., apartment, house, parking)
 - city, region
 - price, created_at, updated_at
 - Agent_id
- Leads / Contacts
 - contact_id
 - listing_id
 - contact_source (organic, paid, partner)
 - Contact_timestamp

(Please generate mock data - the goal is your reasoning, not data volume.)

Your Tasks

- I. Load the Data
 - Assume the data arrives daily in S3 or local CSVs.
 - Describe (or briefly show) how you would ingest it into Snowflake — for example via COPY INTO, Snowpipe, or another approach.
- II. Model with dbt
 - Create at least:
 - A staging model to clean and normalize data.

- A mart model that calculates one key business metric, such as:
“Leads per Active Listing” by property type and region.
 - Use dbt tests and documentation (*schema.yml*) where relevant.
- III. Ensure Data Quality
 - Add 2-3 dbt tests of your choice (e.g., uniqueness, relationships, freshness).
 - Explain why each test matters and what business or technical risk it addresses.
- IV. Show Business Value
 - Demonstrate how your model helps SeLogger: e.g., identify high-conversion regions, detect under-performing listings, or optimize marketing spend.
- V. Document and Explain
 - Add a short README.md, 1-pager doc or slide covering:
 - Architecture choices (AWS → Snowflake → dbt).
 - Key design decisions and trade-offs.
 - How you’d productionize like and monitor this pipeline in a real platform.

Real-World Considerations (please think through these)

While your data may be simple, the *real* world isn’t. As you design your solution, briefly consider or mention how you would handle:

- **Schema drift** – tomorrow, new fields might appear in the listings feed (e.g., *furnished_flag*). How would your models adapt?
- **Slowly changing attributes** – property price or region can change. Would you snapshot or overwrite?
- **Late-arriving leads** – some leads arrive hours or days after the listing update. How would you keep metrics consistent?
- **Data contracts** – what minimal validation rules should upstream systems enforce before landing data in Snowflake?
- **Performance vs. cost** – would you materialize your mart model as a table, view, or incremental table, and why?

(These questions don’t require implementation – they reveal your engineering judgment and sense of scale.)

3. Deliverables

Please provide:

- A gitHub repo:
 - dbt project (models, schema.yml, README).
 - Small CSVs with the data that you generated.
 - Optional: dashboard / SQL query showing your metric.

- The short design note or slide (1–2 pages max).

(Expected total effort: **3–4 hours**.)

Depth and clarity matter more than completeness.)

4. Panel Presentation (60 minutes)

Segment	Duration	Focus
1. Introduction	5 min	Candidate summarizes context & assumptions.
2. Walkthrough	15 min	Present dbt models, architecture, and tests.
3. Business Insight	10 min	Show and interpret the chosen metric.
4. Deep-Dive Q&A	20 min	Discuss design choices, scaling, ingestion, and real-world considerations.
5. Reflection	5 min	What would you improve or extend next time?

5. Tips for Candidates

We don't expect a production-ready pipeline: focus on clear reasoning and structure.

It's perfectly fine to simplify or simulate parts of the setup.

The goal is to show how you think like a data engineer who delivers business value.